policy, the agent would not even be able to move forward, let alone learn to jump over obstacles.

Figure 8.21. The two main approaches for solving the sparse reward problem with curiosity-like methods are prediction error methods, like the one we used in this chapter, and empowerment methods. Rather than trying to maximize the prediction error between a given state and the next predicted state, empowerment methods aim to maximize the mutual information (MI) between the agent's actions and the next states. If the MI between the agent's action and the next state is high, that means the agent has a high level of control (or power) over the resulting next states (i.e., if you know which action the agent took, you can predict the next state well). This incentivizes the agent to learn how to maximally control the environment.

Figure 8.22. Claude Shannon developed communication theory, which was born from the need to encode messages efficiently and robustly across noisy communication channels as depicted here. The goal is to encode the message such that the mutual information between the received message and the sent message is maximal.

Chapter 9. Multi-agent reinforcement learning

Figure 9.1. In the multi-agent setting, each agent's actions not only affect the evolution of the environment, but also the policies of other agents, leading to highly dynamic agent interactions. The environment will produce a state and reward, which each agent 1 through j use to take actions using their own policies. However, each agent's policy will affect all the other agents' policies.

Figure 9.2. In independent Q-learning, an agent does not directly perceive the actions of other agents but rather pretends they are part of the environment. This is an approximation that loses the convergence guarantees that Q-learning has in the single-agent setting, since the other agents make the environment nonstationary.

Figure 9.3. In a stationary environment, the expected (i.e., average) value over time for a given state will remain constant (stationary). Any particular state transition may have a stochastic component, hence the noisy-looking time series, but the mean of the time series is constant. In